

TBA4245 GEODESY

Example exercise no. 2: The Geodesic. Clairaut's equations

This is a task from exam in 2004. Questions (c) and (d) are added in 2007.

- (a) Geodesic
Define the geodesic. Figure.
Substantiate why we use a geodesic.

- (b)
NTNU has a co-operation with the University in Perth (west in Australia).
The distance and the azimuth between Trondheim (point T in the table) and Perth (point P) shall be computed. In the table is point E a point on Equator. All the points are on the surface of the spheroid.
Geographical coordinates and UTM values given in the WGS84 datum:

Point	Latitude, B	Longitude, L	North, N	East, E
T	63.414993056° N	10° 24' 22.8768" E Gr.w.	7032598.60	570218.12
P	31.966666667° S	115° 49' 0.0000" E Gr.w	3537352.32	388183.90
E	0°	97° 41' 31.8700" E Gr.w	0.00	354460.33

Azimuth and distance of the geodesic going through T and P:

From - to	Azimuth, α	Distance
T - P	92.306606032°	13889439.648

Azimuth and distance of the geodesic going through E and P:

From - to	Azimuth, α	Distance
E - P	153.361213135°	4019638.373

Is point E on the geodesic trough the points T and P? Substantiate your answer with computations.

Extra in 2007:

- (c) Calculate the azimuth from P to T.
(d) Calculate the latitude of the northernmost point of the geodesic going through T and P.

Solution, example exercise no 2

Task 2: Geodesic etc.

(a) Geodesic

Define the geodesic. Figure.

Headwords, see chapter 1 in Geodesy part 2" for more: Osculating plane, geodetic curvature, normal sections will not give an unique trace on the spheroid between two points. Geodetic curvature is zero along the geodesic.

Substantiate why we use a geodesic.

Headwords: Unique trace, shortest trace (distance), give Ok formulas to be used for calculations on the spheroid (the geodetic main tasks) and Ok formulas to develop the equations to and from the map projection plane.

(b)

Is point E on the geodesic trough the points T and P? Substantiate your answer with computations.

Using the global datum WGS84 and the spheroid defined there. You will find the values of the semi axis a and b in "Geodesy part 2" and other places.

Clairaut's 2 nd formula of the geodesic: $\cos \beta \cdot \sin A = k = \text{constant}$

Reduced latitudes in the 3 points:

$$\tan \beta = \frac{b}{a} \cdot \tan B \quad \text{gives: } \beta_T = 63,33790552^\circ \text{ N} \quad \beta_E = 0^\circ \quad \beta_P = 31,880304601^\circ \text{ S}$$

The constant k is calculated from given values in point T:

$$k = 0,448364290505. \text{ NB: } k \text{ value is valid for going southwards from T to P.}$$

The constant k is calculated from given values in point E:

$$k = 0,448364290506. \text{ NB: } k \text{ value is valid for going southwards from E to P.}$$

=> The same numerical value of k , the 3 points must be on the same geodesic.

(c)

Using Clairaut's equation:

From T: Calculating the azimuth in P of the geodesic, the value is valid in the same direction as in T, southwards: $\alpha = 148,128675639^\circ$

From P towards T, adding 180 degrees: $\alpha = 148,128675639 + 180 = 328,128675639^\circ$

Control from E: Calculating the azimuth in P of the geodesic, the value is valid in the same direction as in E, southwards: $\alpha = 148,128675639^\circ$ and then adding 180° , Ok.

(d)

The northernmost point on the geodesic, using the value of the azimuth in the northernmost point must be 90 degrees and entering this into Clairaut's equation, the latitude is the only unknown, as the constant k is the same as above:

$$\beta_0 = 63,36121314 \text{ degrees} \quad \Rightarrow \text{Geographical latitude: } B_0 = 63,43825385 \text{ degrees}$$